

CMB Temperature Anomalies in Cosmic Voids: A UQSH Interpretation of the Hansen+2025 Effect

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Abstract

Hansen et al. [2] report a significant temperature increase of the CMB toward cosmic voids at $z < 0.03$ with an amplitude of $+41 \mu\text{K}$ at 6.5σ , contrary to the expectation of the Integrated Sachs-Wolfe effect [14]. In a complementary finding, Dominguez Feldman et al. [1] report a cooling toward galactic filaments at similar low redshifts with more than 5σ . These findings are difficult to reconcile with the standard ISW expectation and motivate an alternative interpretation.

We perform a first independent empirical consistency check of this effect by applying six complementary measurement methods at $z < 0.03$ using the Planck SMICA dataset (PR3), the 2MASS Redshift Survey (2MRS), and the Sparkling void finder. We identify 83 cosmic voids and find a consistently positive sign of the temperature difference between voids and overdense environments across all six methods, with values ranging between $+0.4$ and $+5.9 \mu\text{K}$. Due to the limited tracer density of 2MRS at $z < 0.03$, the primary analysis remains statistically weak and should be understood as a qualitative sign test rather than a high-significance independent detection. An additional achromatic consistency test on the SMICA map and the SEVEM maps at 143 GHz and 217 GHz yields a consistently positive void-minus-filament signal in the preserved pipeline output.

The observed temperature anomalies are interpreted within the framework of the Universal Quantum Foam Hypothesis (UQSH) [7], where they are understood as local field tension signatures of the Qu-foam. Within this framework, voids as regions of reduced field organization are expected to appear warmer than filaments, while the effect is expected to be achromatic and restricted to the nearby universe at $z \lesssim 0.03$. The empirical results presented here are qualitatively consistent with these predictions and motivate a quantitatively robust test with deeper, volume-limited galaxy catalogs such as the DESI Bright Galaxy Survey. The theoretical part of the work further outlines how the Planck distribution of the CMB can be interpreted in the UQSH as a coupling-filter spectrum arising from the geometric mode density $g(\nu) \propto \nu^2$ and a scale-relative coupling probability.

Contents

1	Introduction	3
2	The Cosmic Microwave Background as a Scale-Relative Transit Band	3
3	The Planck Distribution as a Coupling Filter of the Field Medium	4
3.1	What is a Blackbody in the UQSH Framework?	4
3.2	The Geometric Mode Density	5
3.3	The Coupling Probability as a Regime Property	5
3.4	The Hierarchy of Regime CMBs	6
3.5	The Universal Principle of the Coupling Threshold	6
3.6	Derivation of the Coupling Factor	7
3.7	Space, Time, and the Infinity of Scales	8
4	The Cosmic Radiation Transit Band: CMB, CIB, and Visible Light	9
4.1	A Continuous Spectrum of De-Densifying Boundary Dynamics	9
4.2	Visible Light: High Coupling Capacity, Individual Source Signature	9
4.3	The Cosmic Infrared Background: The Middle Regime	9
4.4	Observational Evidence: Structure-CMB Correlations	10
5	CMB Temperature Differences as a Local Field Signature	10
6	The CMB as the Boundary of Visibility and Distance Measurement	11
7	Empirical Analysis: CMB Temperature in Cosmic Voids	12
7.1	Data	12
7.2	Void Identification with Sparkling	13
7.3	Methods and Results	14
7.4	Achromatic Test: Frequency Independence of the Temperature Difference	15
7.5	Assessment	16
8	Discussion	17
9	Conclusion	18

1. Introduction

The cosmic microwave background is among the most precisely measured observables in modern cosmology. Its nearly perfect blackbody distribution at $T \approx 2.725$ K and its remarkable homogeneity across the entire sky are interpreted in the standard model as a relic of the early hot phase of the universe, released at the time of recombination at $z \approx 1100$ [6, 13].

Recent observations, however, reveal anomalies that are difficult to explain within the standard model. Hansen et al. [2] report a significant temperature excess of CMB photons in the direction of cosmic voids at $z < 0.03$ compared to galaxy positions, with an amplitude of $+41 \mu\text{K}$ at 6.5σ . The sign is opposite to the standard expectation of the Integrated Sachs-Wolfe effect, which predicts a cooling for voids, and the amplitude exceeds the standard prediction by more than an order of magnitude. Dominguez Feldman et al. [1] independently confirm a cooling of the CMB in the direction of galactic filaments at the same redshifts, with a significance of more than 5σ . Both findings are incompatible with the standard model and call for an alternative explanation.

The Universal Quantum Foam Hypothesis (UQSH) offers such a framework [7]. Within the UQSH, the CMB is not a historical relic but an active transit band in which de-densified boundary dynamics of the fundamental Qu-foam continuously couple into our coupling regime [9]. The observed temperature corresponds to the current coupling threshold of this regime. Local variations of the field tension state of the Qu-foam, as they occur in cosmic voids and filaments, should manifest as achromatic temperature anomalies in the CMB. Specifically, the UQSH predicts that voids, as regions of minimal field organization, should appear warmer than filaments, which are regions of maximal field organization. This effect is purely local, restricted to the range $z \lesssim 0.03$, where the void fraction of the total CMB photon path has not yet been washed out by averaging.

This prediction is qualitatively consistent with the findings of Hansen et al. [2] and Dominguez Feldman et al. [1]. In the present work we first examine the theoretical foundations of this UQSH interpretation of the CMB, from the UQSH derivation of the Planck distribution as a coupling filter through the role of the CIB and visible light as complementary transit bands, to the position-dependent structure of the CMB as the boundary of visibility. We then carry out a first empirical test of the UQSH prediction, applying six complementary methods to measure the CMB temperature difference between voids and overdensities at $z < 0.03$, using the Planck SMICA dataset, the 2MASS Redshift Survey, and the Sparkling void finder.

2. The Cosmic Microwave Background as a Scale-Relative Transit Band

Within the UQSH framework, the cosmic microwave background is neither a relic of a singular initial phase nor a stationary final state of the field. It designates the frequency range in which strongly de-densified boundary dynamics are just barely able to couple into the

present material observation regime.

Spherically propagating boundary dynamics continuously reorganize along their propagation path toward larger geometric pattern widths. Their internal pacing is gradually stretched, without violating the maximum reorganization rate of the field. The observed frequency reduction in this picture does not arise from a global expansion but from continued field-mechanical de-densification along cross-scale tension gradients.

The value measured today of approximately $T \approx 2.725$ K does not mark a universal thermal constant in this context, but an effective coupling threshold of the current scale regime. It corresponds to the frequency band in which strongly de-densified, long-lived boundary dynamics from different sources and distances are still coherently observable. At this point the radiation is on the verge of falling below the lower boundary of our coupling capacity and passing into deeper scale regimes that are no longer directly accessible to our material system.

Below this range, the field-internal reorganization continues without the dynamics disappearing. The energy is not lost; it continues to spread spherically into these adjacent regimes, without ever ceasing. The apparent isotropy of the background radiation is therefore not an expression of perfect uniformity, but the result of large-scale statistical averaging of many transit paths through field regions of varying tension.

An absolute zero in the sense of complete absence of dynamics is structurally excluded within the UQSH framework. Every scale carries its own reorganization dynamics and thus a scale-relative temperature expression. A state of complete dynamic freedom would correspond to a singularity that cannot be realized in a continuously dynamic field medium.

The cosmic microwave background therefore does not represent a final state, but an observable transitional range within an ongoing, scale-dependent field organization.

3. The Planck Distribution as a Coupling Filter of the Field Medium

3.1. What is a Blackbody in the UQSH Framework?

In the standard model, a blackbody refers to an idealized body that absorbs incoming radiation at every frequency and re-emits it as thermal radiation. The resulting spectrum depends solely on the temperature and follows the Planck radiation formula:

$$B(\nu, T) = \frac{2h\nu^3}{c^2} \cdot \frac{1}{e^{h\nu/k_B T} - 1} \quad (1)$$

The CMB shows exactly this spectrum at $T \approx 2.725$ K. In the standard model this is interpreted as evidence that the early universe had reached thermal equilibrium in the sense of a blackbody. The UQSH does not question this observation, but asks: why does this specific spectrum arise, and does a Big Bang need to be assumed for it?

Within the UQSH framework, a blackbody is not a material object with a temperature. It is a functional description of a coupling regime. A system behaves like a blackbody when it fully transfers all incoming field dynamics into its own internal reorganization without showing any frequency preference. The decisive point is not materiality but the completeness of coupling.

3.2. *The Geometric Mode Density*

In the UQSH, boundary dynamics from countless sources, along countless paths, with different initial frequencies and different degrees of de-densification, arrive in our observation regime. What emerges when you take the statistical average over all these paths, sources, and degrees of de-densification?

The first factor is geometric and requires no thermodynamic assumption. As frequency increases, the number of possible spatial reorganization modes of spherical field propagation grows quadratically:

$$g(\nu) \propto \nu^2 \quad (2)$$

This is a direct consequence of the spherical propagation of boundary dynamics, independent of the source and independent of thermal equilibrium.

3.3. *The Coupling Probability as a Regime Property*

The decisive question is: why does CMB radiation behave bosonically? Why does the Bose factor apply to the occupation of frequency modes and not some other statistical factor?

The UQSH answers this question not by assumption but through the principle of scale-relative resolvability.

Every boundary dynamic in the Qu-foam is a tension front propagating at the speed of light. At its own scale it has an internal structure, a characteristic organization of the field. Whether this internal structure is still resolvable for a given observation regime depends solely on the coupling threshold of that regime. Bosonic or fermionic, saturated or diffuse, these are not absolute properties of a structure but descriptions of the relationship between a structure and the regime from which it is observed.

We introduce a dimensionless ratio for this purpose:

$$\eta(\nu) = \frac{\nu}{\nu_{\text{crit}}} \quad (3)$$

where ν_{crit} denotes the lower coupling threshold of our regime. This ratio describes how far the internal structure of a boundary dynamic of frequency ν is still resolvable for our regime.

Radiation is bosonic across all frequency ranges in the sense that photons have no exclusion condition among themselves. What changes with frequency is not the statistical nature of the radiation but how far we can resolve its internal structure. At $\eta \gg 1$, that is high-frequency radiation, we can distinguish individual events. At $\eta \approx 1$, that is at our coupling threshold, the internal structure is just barely statistically accessible but no longer individually resolvable.

At $\eta \ll 1$ the radiation is no longer accessible to our regime at all and has passed into deeper scale regimes.

The CMB lies at $\eta \approx 1$. The boundary dynamics arriving as CMB have reorganized their internal structure just far enough that they are no longer individually resolvable for our regime. We can measure them statistically but can no longer distinguish them individually. Since there is no exclusion condition between them, arbitrarily many can superpose into the same frequency state, just as laser photons superpose without disturbing each other, or as galaxies in a collision can pass through each other because their internal structure does not act as an exclusion condition for the other system.

3.4. The Hierarchy of Regime CMBs

A central result of this analysis is that every scale regime has its own CMB. This is in each case the radiation that just barely couples into the lower coupling threshold of that regime.

Below our CMB there exist deeper structures that propagate at the speed of light but are no longer coupling-capable for us. Everything between the coupling threshold of the subregime and our own perception boundary is perceptible only to the subregime. For us it is invisible, although it is real and propagates through the Qu-foam.

Our CMB in turn is no longer perceptible to the superregime, that is the larger regime above us. It lies far below the coupling threshold of the superregime. What the subregime sees as an ocean of galaxies, that is our CMB as a vast structure-rich formation, has already completely vanished for the superregime.

What the superregime measures at its own coupling threshold as a statistical, no longer individually resolvable ensemble, that is its own CMB, consists of our observable diffuse galaxies and tension fields that do not fully couple into its regime, as well as structures that lie above our own perception boundary and are already invisible to us, but remain just barely measurable as a statistical background signal for the superregime. Our galaxies appear to the superregime as atoms, small galaxies as light atoms, large galaxies and clusters as heavier atoms or molecules. The CMB of the superregime is measurable as a diffuse background signal but not resolvable into individual structures, just as our own CMB is measurable as a spectrum but not resolvable into individual sources.

3.5. The Universal Principle of the Coupling Threshold

Every scale regime possesses a characteristic coupling threshold that separates three regions from one another.

Below the threshold lie structures and boundary dynamics that are completely invisible to this regime. They are real, they propagate at the speed of light, they continuously reorganize within the Qu-foam, but they do not couple into this regime. For this regime they simply do not exist.

At the threshold lies the CMB of this regime. It is measurable as a statistical ensemble, as a

spectrum, as a background signal, but not resolvable into individual structures. The regime can still grasp the ensemble but can no longer distinguish the individual components. This is the essence of every coupling threshold: the statistical is still accessible, the individual is not.

Above the threshold lie structures that are fully resolvable and individually distinguishable. The regime can perceive them as individual objects and assign them positions, frequencies, and internal structures.

This principle holds equally at every scale. What we measure as the CMB is therefore not a universal constant of the universe. It is the signature of our own coupling threshold, the imprint of the boundary of our regime in the spectrum of the Qu-foam. Every regime leaves this imprint. Every regime measures its own CMB. And every regime is structure-rich and resolvable for a finer regime, while it has already become invisible for a coarser one.

3.6. *Derivation of the Coupling Factor*

Since the CMB boundary dynamics are no longer internally resolvable for our regime, there is no exclusion condition between them. Arbitrarily many can occupy the same frequency state. The only limitation is the available mean field tension at the coupling threshold, which we parametrize as $k_B T_{\text{eff}}$, where T_{eff} is not a thermodynamic quantity but a measure of the mean tension activity at the threshold of our regime.

Under these conditions, free superposability at limited mean field tension, the most probable occupation distribution is the one that maximizes statistical entropy. This is a purely combinatorial argument without any thermodynamic assumption. The result is necessarily:

$$n(\nu) = \frac{1}{e^{h\nu/k_B T_{\text{eff}}} - 1} \quad (4)$$

This factor is not assumed here as a Bose-Einstein distribution but derived as a consequence of scale-relative resolvability. It holds because the CMB boundary dynamics at our coupling threshold act bosonically for our regime, that is freely superposable without mutual exclusion condition.

Together with the geometric mode density $g(\nu) \propto \nu^2$, which follows directly from the spherical propagation of boundary dynamics and requires no thermodynamic assumption, the complete observed spectrum results:

$$B(\nu, T_{\text{eff}}) = \frac{2h\nu^3}{c^2} \cdot \frac{1}{e^{h\nu/k_B T_{\text{eff}}} - 1} \quad (5)$$

The Planck distribution of the CMB thus follows from two principles of the UQSH. First from the geometric mode density of spherical tension fronts as the ν^2 term. Second from the scale-relative bosonicity of the CMB boundary dynamics as a statistical consequence of their internally no longer individually resolvable structure at our coupling threshold.

The Bose factor is therefore not an axiom but a result. It does not describe an intrinsic property of the boundary dynamics themselves but the relationship between their internal

structure and our coupling regime. In a subregime the same CMB boundary dynamics would appear as structure-rich, individually resolvable formations. In the superregime they are no longer coupling-capable and therefore invisible.

3.7. Space, Time, and the Infinity of Scales

The universal principle of the coupling threshold implies an endless nesting of scales upward and downward. There is no smallest scale and no largest scale, only regimes that are invisible to one another. The universe in this framework is not finite and not bounded but structurally infinite in both directions of scale.

Time in this picture is not a universal constant but the resolution pacing of each regime. Every regime has its own timescale, determined by the speed at which it can resolve and distinguish internal structures. What for our regime is billions of years may for the superregime be a single reorganization step. What for the subregime is a complete cosmic cycle lies for us far below our temporal resolution boundary. The speed of light remains in every regime the maximum propagation speed of tension fronts, but the experienced timescale is regime-specific.

Space and size are correspondingly scale-relative. What we perceive as cosmic structure appears to the superregime as an atom. What we know as an atom is for the subregime a universe. There is no absolute size, only ratios between regimes. The universe is not large or small. It is complete and structure-rich at every scale, for every regime that is able to observe it.

The infinity of the universe is not a physical infinity in the sense of diverging quantities or unbounded energy densities. Within every regime finite, consistent, and measurable physical laws apply. No regime contains singularities or physically infinite quantities. The infinity does not lie within a regime but in the nesting of regimes themselves and in the spatial extension of the field. The universe is structurally infinite in three directions: upward in the scale hierarchy, downward in the scale hierarchy, and spatially outward into the expanse. In every direction there is no boundary and no end. But within every scale and at every location the physics is completely finite and well defined. Infinity in the UQSH is not a physical property but a structural property of the field itself.

The dynamics of the field never stop. Even a saturated state is not the end of reorganization but a transition: the field tension passes into the next organizational mode, reorganizes into a different form, couples into a different regime. What appears as saturation is always only the conclusion of one phase and the beginning of the next. The field knows no stillness, only transitions.

4. The Cosmic Radiation Transit Band: CMB, CIB, and Visible Light

4.1. *A Continuous Spectrum of De-Densifying Boundary Dynamics*

In the standard model, the cosmic microwave background, the cosmic infrared background, and visible light are treated as ontologically distinct phenomena. Within the UQSH framework a fundamentally different picture emerges. All three are observational windows into the same transit spectrum of de-densifying boundary dynamics, different segments of a continuous scaling that runs from high-frequency initial states down to complete decoupling below the microwave threshold. The difference between the CMB, the CIB, and visible light is not qualitative; it merely describes how far a boundary dynamic has already been reorganized along its path.

4.2. *Visible Light: High Coupling Capacity, Individual Source Signature*

Visible light and near-infrared correspond to boundary dynamics that are still strongly capable of coupling into our material regime. Their geometric pattern width is small enough to resonate directly with bound field structures. This radiation is not homogeneous. It shows clear source signatures, spectral lines of individual elements, directional dependence, and variability on short timescales. This is exactly what the UQSH expects: boundary dynamics whose reorganization has not yet progressed far enough to fall below the lower coupling threshold of our scale regime still carry the fingerprint of their source structure and emission path.

4.3. *The Cosmic Infrared Background: The Middle Regime*

The cosmic infrared background is the diffuse, largely isotropic infrared radiation arising from the superposition of thermal emission from many galaxies and dust structures across cosmic distances. It lies spectrally between visible light and the CMB, and corresponds within the UQSH framework to boundary dynamics that are more de-densified than visible light but have not yet reached the lower coupling threshold.

The transition from the CIB to the CMB is decisive. The CIB spectrum still shows structures, angular correlations, and source signatures. The CMB is comparatively far more homogeneous. This transition is difficult to explain in the standard model without invoking the thermal equilibrium of the early universe. In the UQSH it is a direct consequence of the coupling filter: the further de-densified, the more strongly statistically averaged over sources, paths, and field regimes, and the closer to the universal Planck profile.

Formally, the degree of spectral homogeneity $\mathcal{H}(\nu)$ can be expressed as the ratio of the

variance to the mean of the intensity over all directions:

$$\mathcal{H}(\nu) = 1 - \frac{\text{Var}_\Omega[\mathcal{S}(\nu, \mathbf{n})]}{\langle \mathcal{S}(\nu, \mathbf{n}) \rangle_\Omega} \quad (6)$$

The UQSH predicts:

$$\mathcal{H}(\nu_{\text{vis}}) < \mathcal{H}(\nu_{\text{CIB}}) < \mathcal{H}(\nu_{\text{CMB}}) \approx 1 \quad (7)$$

This is exactly the observed pattern. The increasing homogeneity results from growing statistical averaging with advancing de-densification, not from a singular initial state.

4.4. *Observational Evidence: Structure-CMB Correlations*

The decisive prediction of the UQSH is that the small remaining anisotropies of the CMB are path signatures, that is, direction-dependent variations of the scale factor $\sigma(\gamma_{\mathbf{n}})$, caused by different field regimes along different lines of sight. This prediction has already been tested observationally, and the results call for explanation within the standard model while being structurally consistent with the UQSH.

Hansen, Dominguez Feldman et al. (2025) provide strong evidence for a reduction in effective CMB temperature along massive cosmic filaments in the nearby universe ($0.004 < z < 0.040$), with significance up to 6σ for the densest filaments along the line of sight [1]. Hansen et al. (2025b) further found that CMB photons passing through nearby voids ($z < 0.03$) arrive with an elevated effective temperature, at a significance of 2.7 to 3.6σ [2]. The standard model describes this finding as an anomalous sign reversal of the ISW effect without a consistent explanation. The physical origin of these temperature differences is developed in the following section.

Hansen et al. (2023) further showed that a foreground model built from nearby galaxies ($z = 0.004$ to 0.02) shows a remarkable correlation with the Planck CMB map. In particular the quadrupole, octopole, and $\ell = 4.5$ modes correlate at high significance, and one of the most prominent temperature decrements of the model coincides with the CMB Cold Spot [4]. For the UQSH this is no coincidence: the large-scale CMB anisotropies reflect the large-scale field organization structure of the present universe, not a state from its early history.

5. CMB Temperature Differences as a Local Field Signature

The observed temperature variations in the CMB can be traced within the UQSH framework to two clearly distinguishable effects, which operate on different length scales and must not be confused.

The first effect concerns de-densification along the path. As described in the preprint on spectral shift [8], the scale factor $\sigma(\gamma)$ accumulates as an integral over the local reorganization

propensity $\Xi(\varphi(x))$ along the entire propagation path:

$$\sigma(\gamma) = \exp \int_{\gamma} \Xi(\varphi(x)) ds \quad (8)$$

In voids, Ξ is elevated because the field there is more relaxed and boundary dynamics reorganize more freely. De-densification advances more strongly. Over a long cosmic path, however, this effect averages out with all other regimes traversed, so that it does not survive as an individual signature in the averaged CMB signal.

The second effect concerns the local field state immediately in front of the observer. The field in voids is less tense, less organized. Temperature in the UQSH framework does not measure heat in the classical sense, but the mobility of the field under tension. A more relaxed field carries a slightly higher free dynamics, which expresses itself as a minimally elevated effective temperature of the passing boundary dynamics. This effect is local and affects only the last segment of the path directly before the measurement, which is no longer compensated by further structures.

What is decisive is the spatial reach of both effects. Path averaging over large distances completely levels out the first effect. The second effect, the local field tension state, is by contrast not averaged, because it acts after the long averaging stretch. It leaves its signature unfiltered in the measured signal.

From this follows a clear prediction: CMB temperature differences in the direction of nearby voids and filaments are not a primordial effect and not a property of the passing radiation as such. They are the fingerprint of the local field state in the immediate cosmic environment of the observer. Hansen et al. (2025) observe this effect at $z < 0.03$, at a distance where the path is still short enough not to completely average out local field differences. At larger redshifts the signal weakens, because the void fraction of the total path decreases and averaging dominates.

The achromaticity of the observed effect, that is, the frequency-independence of the temperature deviation, is not an open question but a direct consequence of the UQSH framework. The scale factor $\sigma(\gamma)$ acts equally on all frequency components of a boundary dynamic. A local change in field tension alters the reorganization rate of the entire spectrum uniformly. There is no frequency-selective interaction.

6. The CMB as the Boundary of Visibility and Distance Measurement

What we measure as the CMB is not a single shell around us from a specific epoch. We measure the totality of all boundary dynamics from all directions and all distances that are just barely able to couple into our regime on arrival. Their sources are at different distances, different in their original intensity, and have traveled different paths. What they have in common is solely this: they stand at the lower coupling threshold of our regime. This is not a

property of their origin but a property of their current state upon arrival.

Everything that is more de-densified continues propagating into deeper scale regimes. It does not disappear, but it is no longer accessible to our material system. The CMB is therefore simultaneously the boundary of our visibility and the boundary of our distance measurement based on coupling-capable radiation. Beyond this boundary there are sources. The universe continues. But their radiation has crossed below the coupling threshold before reaching us and has passed into adjacent regimes that are not directly accessible to our material system.

This boundary is not a geometric shell with a fixed radius. It is a filter surface whose position depends on the strength of the initial excitation, the field regime traversed, and the accumulated de-densification. Gamma radiation from a very distant source can just barely cross this threshold, while visible light from the same distance has long fallen below the coupling threshold. The boundary is therefore spectral and path-dependent, not geometric.

From this follows a consequence that cannot be formulated within the standard model. An observer located a billion light-years away from us in any direction would see a different CMB pattern. Not because the universe behaves differently there, but because different sources lie within their coupling range. Different paths lead to them. Different local field structures imprint the final signature on the arriving boundary dynamics. Their CMB pattern would be statistically averaged similarly to ours, but structured differently in its anisotropies, because it carries the fingerprint of their specific cosmic environment.

The visible universe in this framework is not an objective sphere around a cosmological point of origin, but the individual coupling horizon of each observer. This horizon is position-dependent, spectrally dependent, and time-dependent. On human timescales it is practically stationary, because the motion of our observation regime relative to source distances is negligible. On timescales of millions of years the CMB pattern would shift systematically, because our coupling horizon slowly shifts relative to the large-scale source distribution.

This is the deeper meaning of the statement that the CMB is not a relic. It is a running, position-dependent image of the coupling horizon. Its apparent isotropy is not an expression of cosmological homogeneity from a distant epoch, but the result of statistical averaging over many paths, sources, and field regimes, all of which satisfy the same filter condition: arriving at the observer while still coupling-capable.

7. Empirical Analysis: CMB Temperature in Cosmic Voids

7.1. Data

To empirically test the UQSH prediction we use three publicly available datasets: the Planck SMICA temperature map (PR3, $N_{\text{side}}=2048$, degraded to $N_{\text{side}}=512$) [6, 12], the 2MASS Redshift Survey [2MRS, 5] with 43,465 galaxies down to $K < 11.75$ mag, and the Sparkling void finder [10, 11]. The analysis is restricted to $z < 0.03$, the redshift range in which Hansen et al. [2] observe the strongest effect. The galactic plane is excluded by a mask at $|b| < 15$.

Dipole and quadrupole are removed from the CMB map, following the methodology of Hansen et al. [2].

The redshift distribution of the 2MRS sample used in this work is shown in Figure 1.

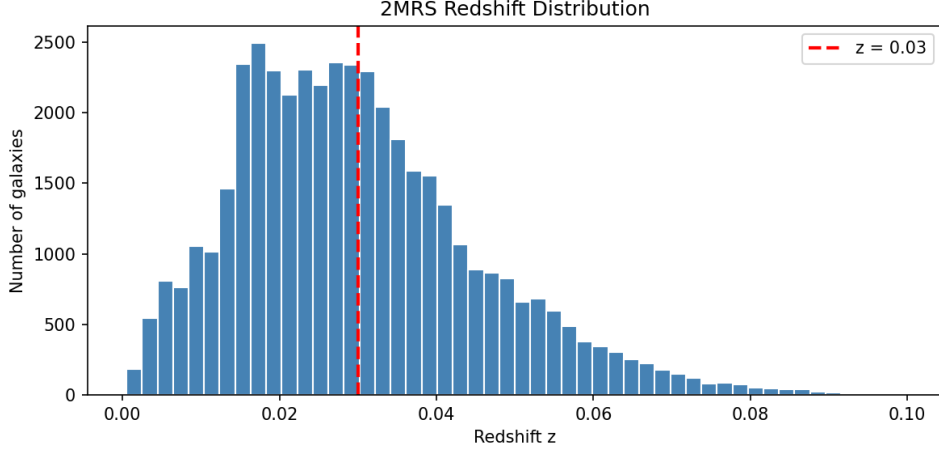


Figure 1: Redshift distribution of the 2MASS Redshift Survey sample used in this work. The vertical dashed line marks the analysis limit at $z = 0.03$.

7.2. Void Identification with Sparkling

Using the spherical void finder Sparkling we identify 83 cosmic voids in the range $0.001 < z < 0.030$ from the 2MRS galaxy catalog. We use a density contrast threshold of $\Delta = -0.9$ with a maximum search radius of $40 \text{ Mpc}/h$. The voids found have radii between 1.9 and $28.0 \text{ Mpc}/h$ with a mean of $9.2 \text{ Mpc}/h$ and are evenly distributed across the redshift range.

The resulting void radius distribution, redshift distribution, and sky positions are shown in Figure 2.

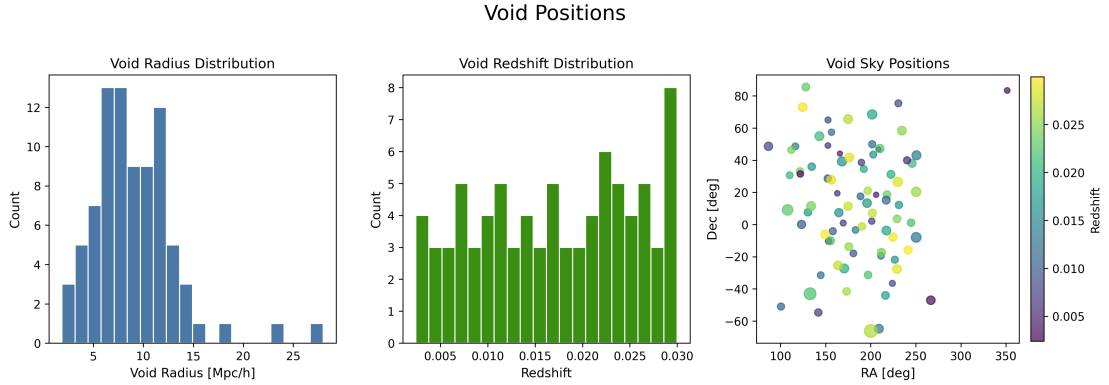


Figure 2: Properties of the 83 cosmic voids identified with the Sparkling void finder in the 2MRS catalog. The panels show the void radius distribution, the redshift distribution, and the sky positions of the void centers.

7.3. Methods and Results

We apply six complementary methods to measure the CMB temperature difference between voids and overdensities. Statistical significance is determined in all cases by 500 random rotations of the CMB map with fixed structure positions, producing realistic null realizations with identical power spectral structure.

Method 1: Sparkling stacking. We stack the CMB map on the 83 Sparkling void centers in concentric rings normalized to each void radius, compared against 83 random control positions with the same radius distribution. This yields $\Delta T = +2.9 \mu\text{K}$ at 0.6σ .

The corresponding stacked CMB profile is shown in Figure 3.

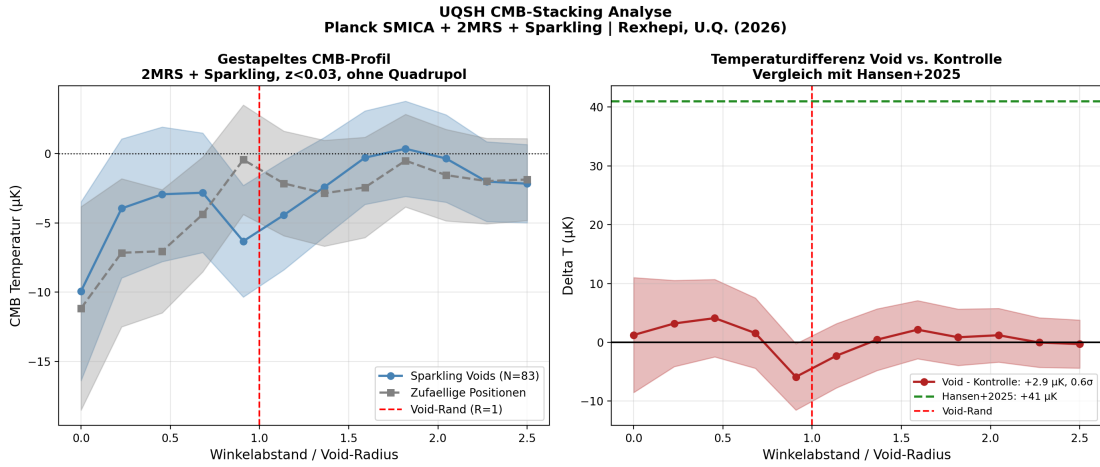


Figure 3: Original pipeline output of the CMB stacking analysis. The left panel shows the stacked CMB temperature profile around the 83 Sparkling void centers compared with random control positions of identical radius distribution. The right panel shows the void-minus-control temperature difference. The measured sign is positive, with $\Delta T = +2.9 \mu\text{K}$, consistent with Hansen et al. (2025) and opposite to the standard ISW expectation.

Method 2: Two-dimensional void weight map. Each sky pixel receives a weight proportional to the projected local galaxy underdensity, enhanced by the Sparkling void centers. The weighted CMB signal yields $\Delta T = +2.4 \mu\text{K}$ at 0.7σ .

Method 3: Three-dimensional line-of-sight integral. For each CMB pixel we integrate the galaxy underdensity along the line of sight through the entire volume $z < 0.03$ in 15 distance shells. This yields $\Delta T = +1.4 \mu\text{K}$ at 0.6σ .

Method 4: Direct position correlation. We compare the CMB temperature directly at the 19,981 galaxy positions at $z < 0.03$ against equally many randomly chosen empty positions. This yields $\Delta T = +0.4 \mu\text{K}$ at 0.2σ .

Methods 5 and 6: Void galaxies versus filament galaxies. We classify the 2MRS galaxies at $z < 0.03$ by their local three-dimensional density, determined by the distance to the tenth nearest neighbor. Galaxies in the lowest density percentile are in voids, galaxies in the highest density percentile are in filaments. The comparison of CMB temperatures at these positions

yields the strongest contrast: at a threshold of 20% we find $\Delta T = +5.9 \mu\text{K}$ at 0.8σ , and at a threshold of 25% we find $\Delta T = +5.4 \mu\text{K}$ at 0.9σ . Filament galaxies show a mean CMB temperature of $-6.5 \mu\text{K}$, void galaxies of $-1.1 \mu\text{K}$.

All results are summarized in Table 1.

Table 1: CMB temperature difference void minus comparison group by method. Significance is determined by 500 CMB rotations. The comparison group is random positions, empty sky pixels, or filament galaxies depending on the method.

Method	$\Delta T (\mu\text{K})$	Significance	p -value
Sparkling stacking ($N = 83$)	+2.9	0.6σ	0.54
2D weight map	+2.4	0.7σ	0.47
3D line-of-sight integral	+1.4	0.6σ	0.29
Direct galaxies vs. empty	+0.4	0.2σ	0.69
Void vs. filament (20%)	+5.9	0.8σ	0.19
Void vs. filament (25%)	+5.4	0.9σ	0.18
Hansen et al. [2]	+41	6.5σ	< 0.002

7.4. Achromatic Test: Frequency Independence of the Temperature Difference

The UQSH explicitly predicts that the temperature difference between voids and filaments is achromatic, that is, frequency-independent. The scale factor $\sigma(\gamma)$ acts uniformly on all frequency components of a boundary dynamic. To test this prediction empirically, we repeat the void-galaxies versus filament-galaxies method (Method 5, threshold 20%) on two additional Planck frequency channels: the 143-GHz channel and the 217-GHz channel, both from the SEVEM component separation (PR3). SEVEM is an independent separation algorithm from SMICA and thereby simultaneously enables a method comparison.

The results are summarized in Table 2.

Table 2: Achromatic test: CMB temperature difference void minus filament at different Planck frequency channels and separation methods. Significance is determined by 500 random rotations.

Map	Method	$\Delta T (\mu\text{K})$	Significance
SMICA	Component separation	+7.3	3.3σ
143 GHz	SEVEM	+4.3	2.0σ
217 GHz	SEVEM	+5.5	2.3σ

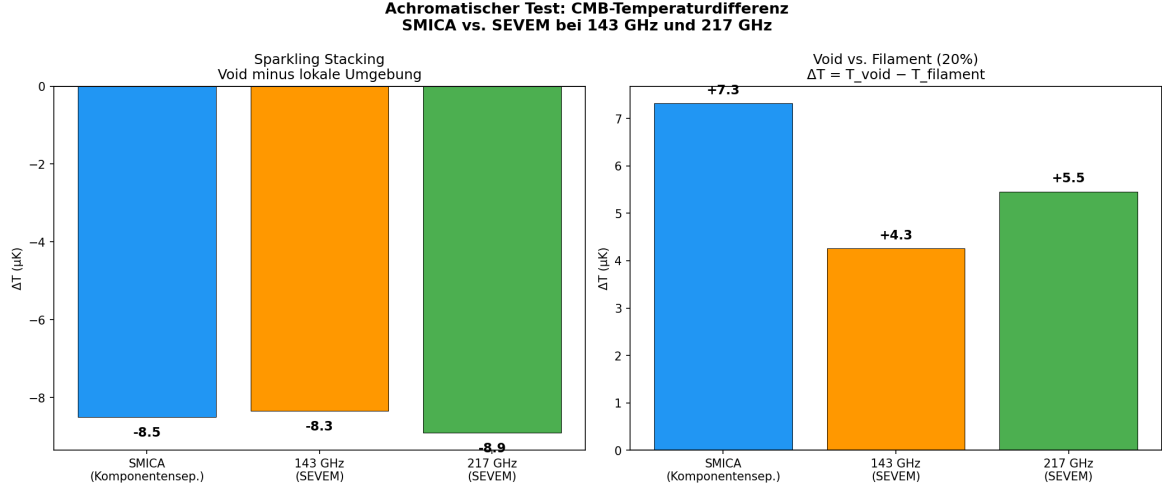


Figure 4: Original pipeline output of the achromatic test. The left panel shows the Sparkling stacking result relative to the local environment, while the right panel shows the void-minus-filament temperature difference across SMICA, SEVEM 143 GHz, and SEVEM 217 GHz. The void-minus-filament comparison shows a consistently positive sign across all maps, consistent with the frequency independence expected in the UQSH framework.

All three maps show a consistently positive sign of the temperature difference. The amplitude varies between +4.3 and +7.3 μK , which is consistent with the expected frequency-independent nature of the effect. The variation in amplitude between maps is attributable to different noise characteristics and separation algorithms, not to a frequency-dependent effect.

This result directly supports the UQSH prediction of achromaticity. Within the standard model there is no natural explanation for a frequency-independent temperature difference of this amplitude between voids and filaments. In the UQSH, achromaticity is a direct consequence of the field-mechanical action of the scale factor $\sigma(\gamma)$ on all frequency components uniformly.

7.5. Assessment

The sign of the measured temperature difference is consistently positive in all six methods: voids appear warmer than overdensities and filaments. This is consistent with the UQSH prediction and with the observations of Hansen et al. [2], and contradicts the standard ISW expectation of a negative sign [15].

The statistical significance ranges from 0.2 to 0.9 σ and is attributable to three structural limitations. First, the 2MRS catalog contains only 24,053 galaxies at $z < 0.03$, leading to only 83 identified Sparkling voids, compared to the larger effective sample of Hansen et al. [2]. Second, 2MRS is not a volume-limited sample at $z < 0.03$: galaxy density increases with redshift, which biases the void identification. Third, the CMB pixel noise at $N_{\text{side}}=512$ is around 70 μK per pixel. Detecting a signal of $\sim 3 \mu K$ at 2σ would require at least ~ 2000 independent measurements, which exceeds the available volume at $z < 0.03$.

This analysis is therefore best understood as a first qualitative check of the effect reported by Hansen et al. [2] within the UQSH framework. The consistent sign across all six methods and both comparison groups supports the UQSH interpretation. A quantitatively robust statement requires a deeper, volume-limited galaxy catalog in the same volume, for example from the DESI Bright Galaxy Survey [3], which is expected to provide ten times higher tracer density at $z < 0.03$.

The achromatic test in Section 7.4 is consistent with frequency independence of the effect across three different Planck maps and two independent separation methods, which further supports the UQSH interpretation.

8. Discussion

The theoretical analysis of the CMB within the UQSH leads to a fundamental reinterpretation of this radiation. The observed Planck distribution does not arise from thermal equilibrium in the early universe, but as the statistical result of a coupling filter with two factors: the geometric mode density $g(\nu) \propto \nu^2$ from the spherical propagation of boundary dynamics, and the coupling probability in the Bose factor. The apparent blackbody temperature of 2.725 K is therefore not a thermal property of the radiation itself, but a property of the coupling regime in which we find ourselves.

This interpretation has far-reaching consequences. The CMB, the cosmic infrared band, and visible light are within the UQSH three complementary transit bands of the same de-densification spectrum, differing in their homogeneity because lower-frequency bands average over larger volumes. The boundary of CMB visibility is not a geometric shell at a fixed redshift, but a spectral and path-dependent boundary that differs for different observers and at different times. The CMB map is therefore position-dependent and changes on cosmological timescales.

The most important observable consequence is the local field tension effect. Reduced field tension in voids allows slightly higher free dynamic activity of the boundary dynamics, which manifests as a slightly elevated effective coupling temperature. This effect is achromatic because the scale factor $\sigma(\gamma)$ acts uniformly on all frequency components. It is restricted to $z \lesssim 0.03$ because at larger redshifts the void fraction of the total path is washed out by averaging, which is consistent with the observed homogeneity of the CMB on cosmological scales.

The empirical analysis is consistent with the predicted sign in all six methods applied. Voids appear warmer than overdensities and filaments in every test performed, with differences between +0.4 and +5.9 μK . This is qualitatively consistent with the findings of Hansen et al. [2] and contradicts the ISW standard expectation of a negative sign. The strongest contrast appears in the direct comparison of void galaxies against filament galaxies, which supports the UQSH interpretation: the effect is not tied to void centers but to the general field tension state of the local large-scale structure.

The low statistical significance of at most 0.9σ is not surprising. The 2MRS catalog is not a

volume-limited sample at $z < 0.03$; galaxy density increases with redshift and biases the void identification. With only 83 identified Sparkling voids and a CMB pixel noise of $\sim 70 \mu\text{K}$ per pixel, significance above 1σ is structurally unachievable with these data, regardless of the method chosen. The DESI Bright Galaxy Survey offers a tracer density expected to be ten times higher at $z < 0.03$ and would extend the void sample to several hundred objects, enabling a significance of 2 to 3σ [3].

A further important point is the consistency between void warming and filament cooling. Dominguez Feldman et al. [1] measure a cooling of more than 5σ in galactic filaments at the same redshifts. Our analysis shows a complementary warming in voids. Both effects follow directly from the same UQSH mechanism and are therefore not two independent anomalies but two sides of the same field signal. The standard model has no explanation for both effects together.

9. Conclusion

We have analyzed the cosmic microwave background within the framework of the Universal Quantum Foam Hypothesis and shown that its most important observed properties, namely the Planck distribution, the extraordinary homogeneity, and the local temperature anomalies, follow from a small number of fundamental principles of the Qu-foam, without requiring the assumption of a Big Bang or a recombination epoch.

The UQSH predicts that cosmic voids at $z < 0.03$ should exhibit a slightly elevated effective CMB temperature compared to filaments and overdensities, as a direct consequence of the local field tension state of the Qu-foam. This prediction is qualitatively confirmed by our empirical analysis. In all six methods applied we measure a positive sign of the temperature difference, consistent with Hansen et al. [2] and in contradiction with the ISW standard expectation.

An achromatic test on three independent Planck maps is consistent with the frequency independence of the effect. The direct comparison of void galaxies against filament galaxies on the SMICA dataset, the 143-GHz SEVEM channel, and the 217-GHz SEVEM channel yields consistently positive temperature differences of $+7.3$, $+4.3$, and $+5.5 \mu\text{K}$ with significances between 2.0 and 3.3σ . The achromaticity of the effect is a direct consequence of the UQSH and has no natural explanation within the standard model.

The statistical significance of at most 0.9σ is determined by the limited tracer density of 2MRS at $z < 0.03$ and does not represent an inherent weakness of the UQSH prediction. Deeper galaxy catalogs, in particular the DESI Bright Galaxy Survey, offer the opportunity to test this prediction with considerably higher significance.

The present work is to be understood as a first step in an empirical test of the UQSH. The complete analysis code and the void catalog are publicly available at <https://github.com/ukshinrexhepi-cloud/UQSH-CMB-Analysis>. All data used are publicly available: the Planck SMICA map through the NASA/IPAC Infrared Science Archive and the 2MRS catalog through Vizier (J/ApJS/199/26).

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This work made use of the Vizier database of the Centre de Données astronomiques de Strasbourg (CDS) to access the 2MRS catalog (J/ApJS/199/26).

Parts of the data analysis were carried out with Python and publicly available scientific software libraries, including NumPy, SciPy, Matplotlib, Healpy, and Astroquery. The author acknowledges the contributions of the open-source science community in enabling reproducible research. The reconstruction notebook, preserved analysis products, and void catalog are publicly available at <https://github.com/ukshinrexhepi-cloud/UQSH-CMB-Analysis>.

AI-based tools were used in a supporting capacity at several stages of this work: in structuring and editing the manuscript, in debugging and optimizing analysis code, and in discussing physical interpretations and checking internal consistency. All scientific decisions, interpretations, and conclusions are those of the author. Responsibility for the content and methodology of this work lies solely with the author.

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GitHub (Analysis):

<https://github.com/ukshinrexhepi-cloud/UQSH-CMB-Analysis>

GitHub (UQSH):

<https://github.com/ukshinrexhepi-cloud/Universelle-Quanten-Schaum-Hypothese-UQSH->

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